



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF ELECTRICAL ENGINEERING

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SKEF2123 INSTRUMENTATION & MEASUREMENT

EB1

Lab Report

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GROUP 4

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Video Link: <https://youtu.be/rwaZ7G1PT8E>

1.0 Introduction

Electrical circuits form the foundation of many electronic and engineering systems. Understanding the relationship between voltage, current, and resistance is essential in circuit analysis. In this experiment, Kirchhoff's Voltage Law (KVL) and Kirchhoff's Current Law (KCL) are applied to analyze a DC resistive circuit, while the Wheatstone bridge is used as a precise method for measuring and verifying resistance values. By combining theoretical analysis with practical circuit construction and measurement, this experiment demonstrates the application of fundamental circuit laws and measurement techniques in real electrical systems.

2.0 Objectives

- To perform current and voltage measurements using electrical and electronic instruments.
- To perform resistance measurement using a Wheatstone bridge.

3.0 Procedures

Part A: Voltage and Current Measurement

- The circuit is connected as shown in Figure 1 by using the provided apparatus.

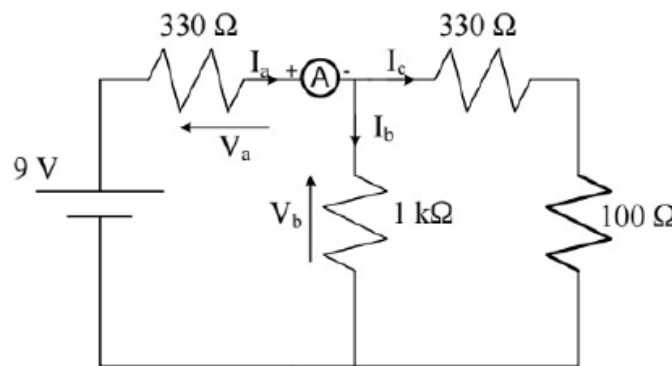


Figure 1: Circuit diagram

- Using Kirchhoff's circuit laws, I_a , I_b , I_c , V_a and V_b of Figure 1 is calculated.
- Appropriate electrical instruments are used to measure I_a , I_b , I_c , V_a and V_b .
- The table is completed for all the parameters.

Part B: Resistance measurements- Wheatstone Bridge

- Given the scale of resistance for potential meter is from 0-10 and the sensitivity of the resistance to scale is 2.5k /unit scale.
 - Graph of the resistance versus knob scale is plotted.
 - The current deflection meter using analog resistance meter when knob at scale 2,4,6,8,10 is measured and percentage of deflection on scale is shown.
- The circuit is connected as shown in figure 2.

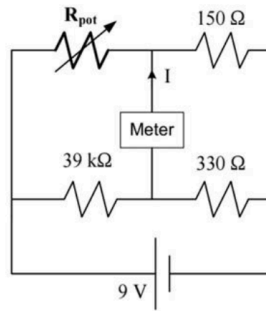


Figure 2: Wheatstone's Bridge

3. The value of the potential meter is calculated when the bridge is balanced.
4. The knob of the potential meter is turned to make the bridge in balance condition. The value of the potential meter is measured and it is recorded as $R_{\text{pot, balanced}}$.
5. By using digital voltmeter and 10 reading is selected at knob resistance at scale 1-10, then the voltages of the bridge are measured using digital meter. Graph voltages versus resistances is plotted.

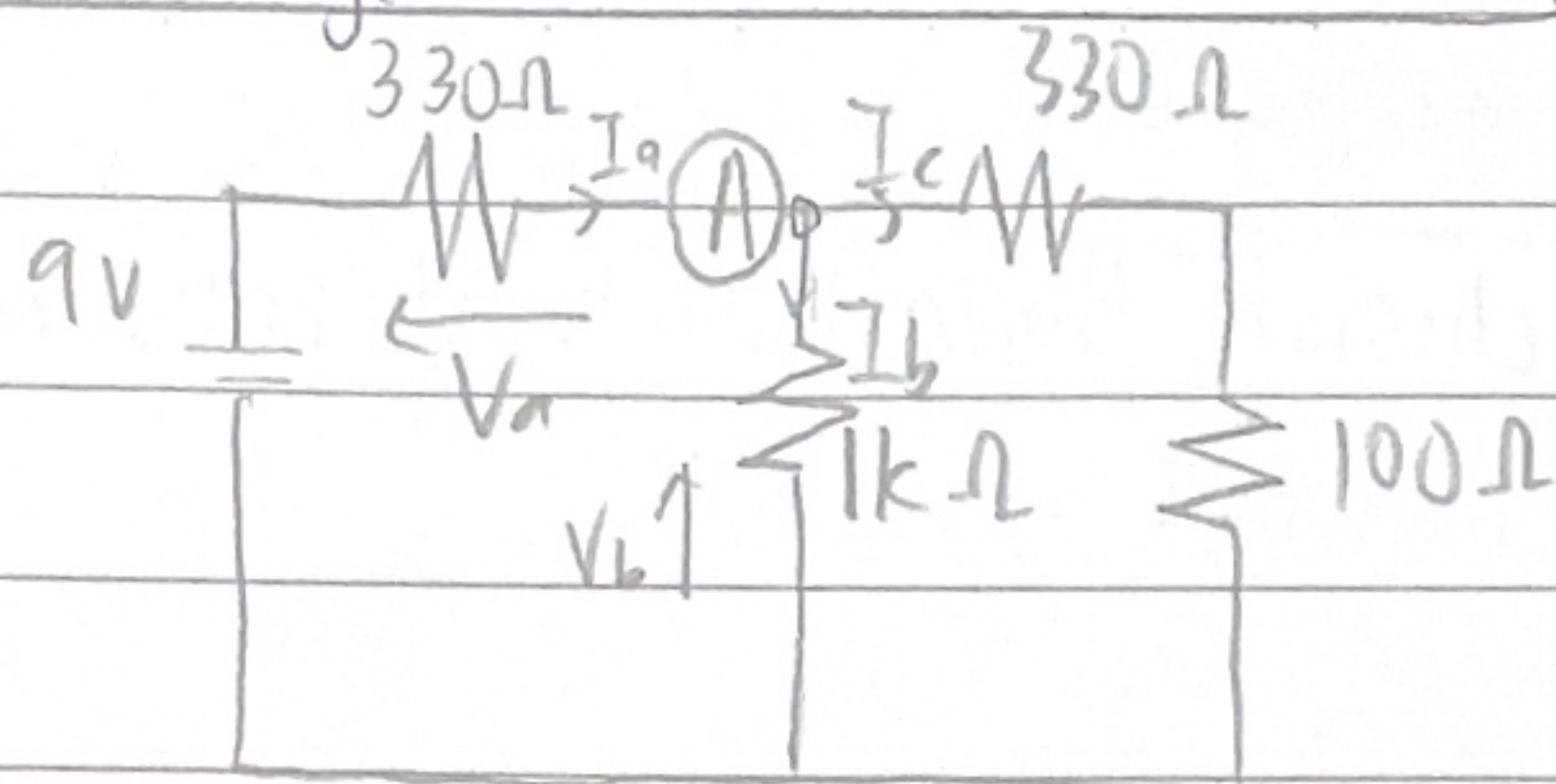
Name: Lam Yoke Yu, Goe Jie Ying, Teh Pu Qian

NO. Title: Electrical Measurement

DATE:

Part A: Voltage and Current Measurement

2.



$$\text{KCL: } I_a = I_b + I_c \quad \text{--- (1)}$$

Sub (1) into (4)

$$\text{KVL: } 9 = 330I_a + 1000I_b \quad \text{--- (2)}$$

$$I_a = 1.43(9.98 \times 10^{-3})$$

$$330I_c + 100I_c - 1000I_b = 0$$

$$I_a = 14.27 \text{ mA} \quad \text{--- (6)}$$

$$430I_c = 1000I_b$$

$$I_b = 0.43I_c \quad \text{--- (3)}$$

Sub (5) into (3)

Sub (3) into (1)

$$I_b = 0.43(9.98 \times 10^{-3})$$

$$I_a = 0.43I_c + I_c$$

$$I_b = 4.29 \text{ mA} \quad \text{,,}$$

$$I_a = 1.43I_c \quad \text{--- (4)}$$

$$V_a = I_a \times R$$

Sub (4) & (3) into (2)

$$V_a = 14.27 \times 10^{-3} \times 330$$

$$9 = 330(1.43I_c) + 1000(0.43I_c)$$

$$V_a = 4.709 \text{ V} \quad \text{,,}$$

$$9 = 471.9I_c + 430I_c$$

$$V_b = I_b \times R$$

$$I_c = 9.98 \text{ mA} \quad \text{--- (5)}$$

$$V_b = 4.29 \times 10^{-3} \times 1000$$

$$V_b = 4.29 \text{ V} \quad \text{,,}$$

$$\therefore I_a = 14.27 \text{ mA} \quad \text{,,}$$

$$I_b = 4.29 \text{ mA} \quad \text{,,}$$

$$I_c = 9.98 \text{ mA} \quad \text{,,}$$

$$V_a = 4.71 \text{ V} \quad \text{,,}$$

$$V_b = 4.29 \text{ V} \quad \text{,,}$$

4.

Parameters	Calculation	Measurement 1 (Analog)	Measurement 2 (Digital)
I_a (mA)	14.27	12.5	11.79
I_b (mA)	4.29	3.4	3.42
I_c (mA)	9.98	8.2	8.08
V_a (V)	4.71	3.9	3.85
V_b (V)	4.29	3.4	3.41

5. All experimentally measured values for analog and digital instruments are consistently lower than the theoretical values. This is because the actual voltage measured is 8.66V which is lower than the 9V used in the calculation.
6. The digital multimeter provides a clear and high-precision values, while analog multimeter is subjective and prone to parallax error, which leads to less precise data.

Part B: Resistance measurements - Wheatstone's Bridge

3.
$$\frac{R_{pot}}{150} = \frac{39000}{330}$$

$$R_{pot} = 17.727 k\Omega$$

4. $R_{pot, balanced} = 17.29 k\Omega$

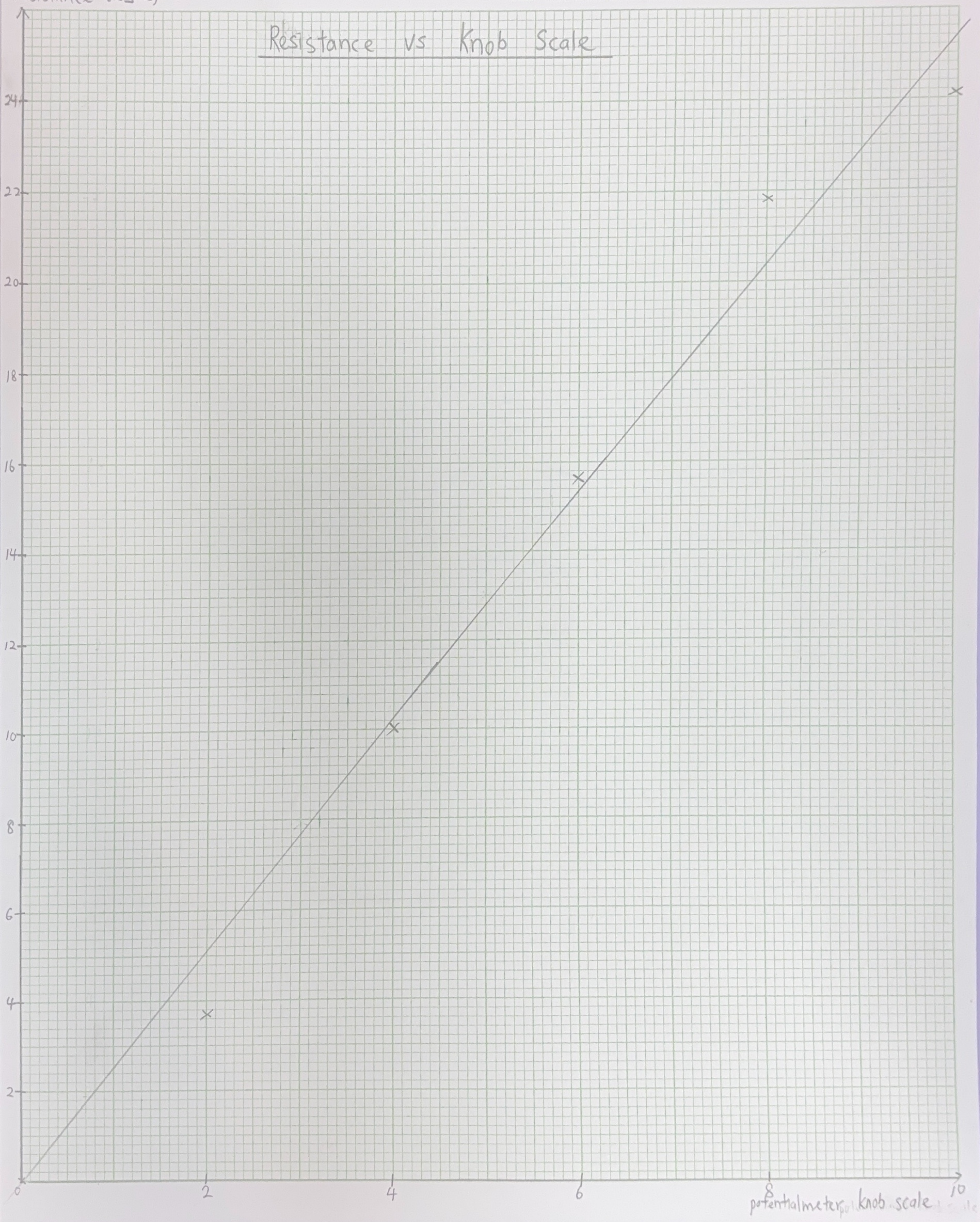
5. The value of $R_{pot, balanced}$ is slightly lower R_{pot} . This because the actual voltage and resistances used are lower than the value used for calculation.

resistance (k Ω)

No.

Date:

Resistance vs Knob Scale



Voltage vs Potential Meter Knob Scale

Voltage of the bridge (V)

No. _____

Date: _____

